

FECAP – CENTRO UNIVERSITÁRIO ÁLVARES PENTEADO

Curso de Ciência da Computação

Disciplina: Sistemas Operacionais e Computação em Nuvem

Professor: Rodrigo da Rosa

Saulo Ribeiro Santos

Gustavo Henrique Da Silva Santos

Luan Rocha da Silva

Thiffany Moraes Vieira da Silva

ASA CONNECT — AGENTE PARA O ESTUDANTE

Relatório Técnico de Infraestrutura — Entrega 1 (Parcial)

Configuração do Ambiente Linux em Contêiner Docker

São Paulo

2026

1 OBJETIVO

Este relatório documenta a configuração e a validação do ambiente operacional Linux, provisionado em contêiner Docker, utilizado como alicerce tecnológico do projeto ASA Connect (Agente para o Estudante), desenvolvido no âmbito do Projeto Interdisciplinar para a Área do Sucesso Alvarista (ASA) da FECAP. Busca-se comprovar a capacidade da equipe de provisionar um ambiente seguro, reproduzível e auditável, aplicando princípios de menor privilégio, gestão de permissões, isolamento de processos, observabilidade de recursos e proteção de segredos, como base para o deploy dos serviços de backend e inferência de IA previstos na Entrega 2.

2 DESENVOLVIMENTO

2.1 Ambiente e Infraestrutura de Contêiner

O ambiente selecionado utiliza contêineres Docker sobre a distribuição Ubuntu 22.04 LTS (Jammy Jellyfish), provisionados via Docker Engine (v29.8.0). Essa escolha justifica-se por três fatores: (i) reprodutibilidade — paridade entre os ambientes locais da equipe e a produção em nuvem (Render/Docker Cloud); (ii) eficiência de recursos — compartilhamento do kernel do host, com inicialização instantânea e baixo consumo de RAM/CPU frente a máquinas virtuais hipervisoras; e (iii) segurança e isolamento — uso de namespaces Linux (PID, NET, IPC, MNT, UTS) e cgroups, isolando o processo do backend do sistema hospedeiro.

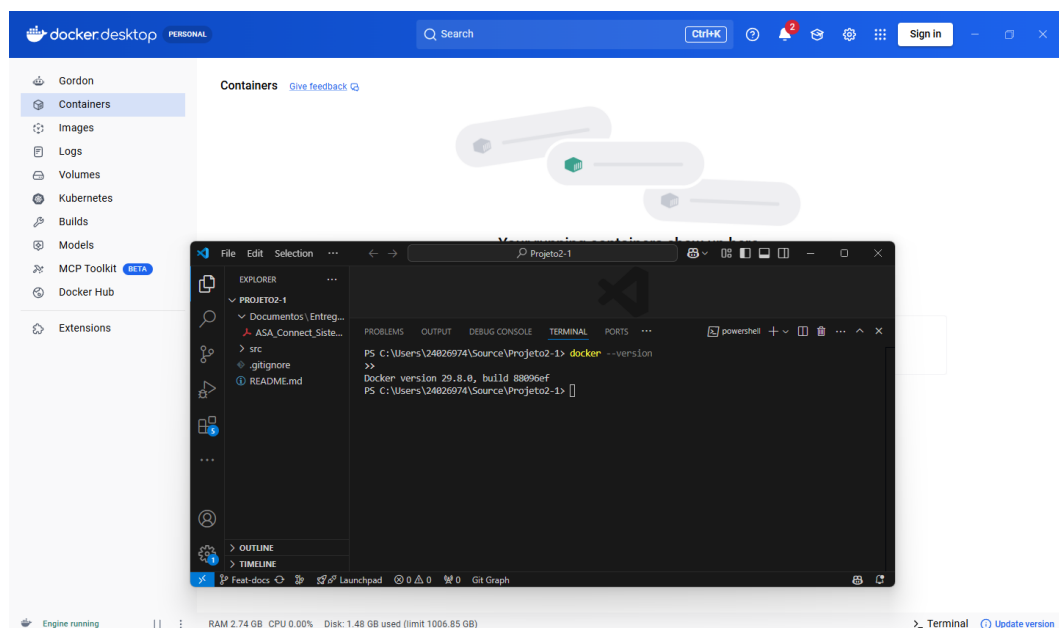


Figura 1 – Docker Desktop exibindo o contêiner ativo do ambiente de infraestrutura.

Fonte: elaborado pelos autores (2026).

2.2 Criação do Contêiner

Comando utilizado para provisionamento e acesso interativo ao contêiner:

```
docker run -it --name asa-connect-infra -p 8000:8000 ubuntu:22.04 bash
```

Parâmetros: `-it` aloca pseudo-TTY com STDIN interativo; `--name asa-connect-infra` define identificador padronizado; `-p 8000:8000` mapeia a porta reservada à API FastAPI; `ubuntu:22.04` é a imagem oficial base; `bash` é o entrypoint do shell.

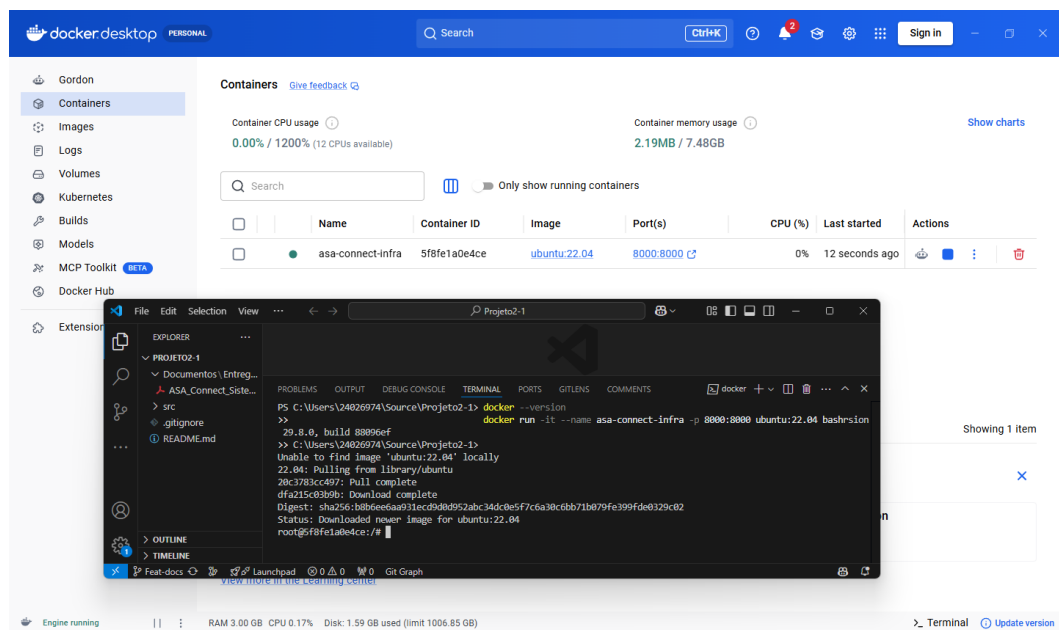


Figura 2 – Execução do comando docker run e obtenção do prompt root@ dentro do contêiner.

Fonte: elaborado pelos autores (2026).

2.3 Atualização do Sistema e Criação de Usuário

Atualização de pacotes e correção de vulnerabilidades conhecidas:

```
apt update && apt upgrade -y
```

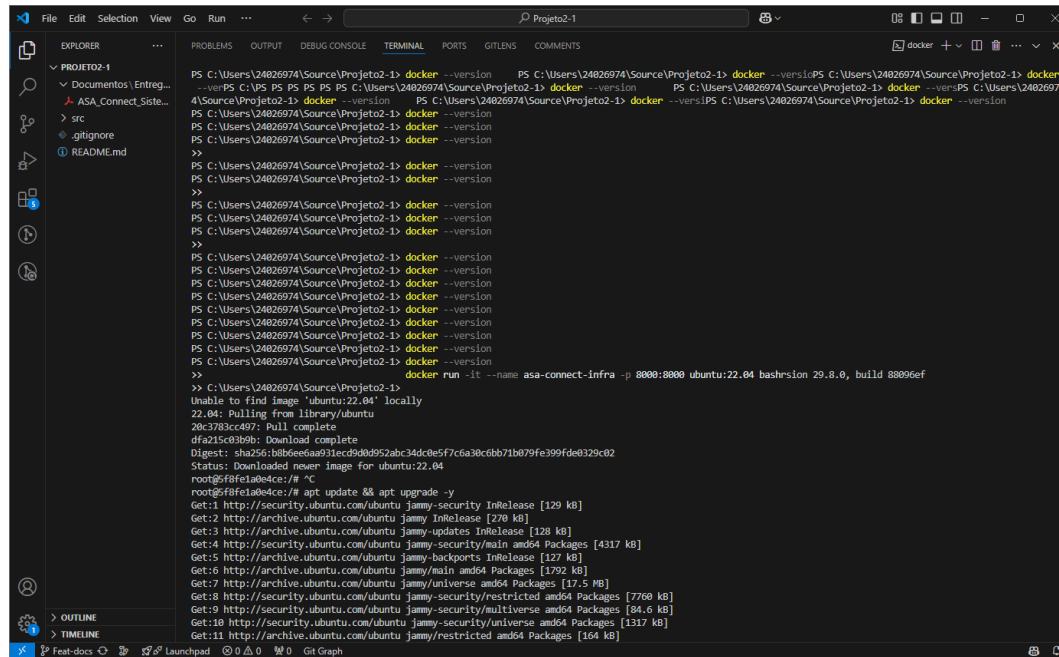


Figura 3 – Execução do apt update && apt upgrade -y no contêiner.

Fonte: elaborado pelos autores (2026).

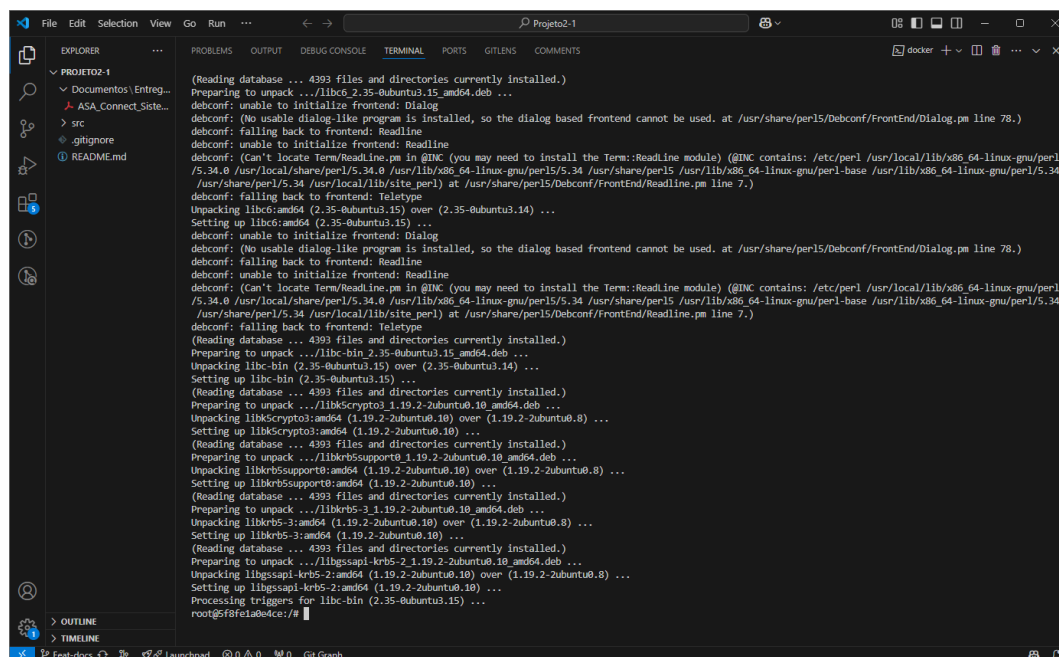


Figura 4 – Conclusão da atualização de pacotes do sistema.

Fonte: elaborado pelos autores (2026).

Em atenção ao Princípio do Menor Privilégio (PoLP), foi criado o usuário não-root asaadmin, evitando a execução de serviços sob root:

```
adduser asaadmin  
usermod -aG sudo asaadmin
```

```

debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC co
ntains: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.34.0 /usr/local/share/perl/5.34.0 /usr/lib/x86_64-l
inux-gnu/perl5/5.34 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.34
/usr/share/perl/5.34 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 7.)
debconf: falling back to frontend: Teletype
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../libc-bin 2.35-0ubuntu3.15 amd64.deb ...
Unpacking libc-bin (2.35-0ubuntu3.15) over (2.35-0ubuntu3.14) ...
Setting up libc-bin (2.35-0ubuntu3.15) ...
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../libk5crypto3 1.19.2-2ubuntu0.10 amd64.deb ...
Unpacking libk5crypto3:amd64 (1.19.2-2ubuntu0.10) over (1.19.2-2ubuntu0.8) ...
Setting up libk5crypto3:amd64 (1.19.2-2ubuntu0.10) ...
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../libkrb5support0 1.19.2-2ubuntu0.10 amd64.deb ...
Unpacking libkrb5support0:amd64 (1.19.2-2ubuntu0.10) over (1.19.2-2ubuntu0.8) ...
Setting up libkrb5support0:amd64 (1.19.2-2ubuntu0.10) ...
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../libkrb5-3 1.19.2-2ubuntu0.10 amd64.deb ...
Unpacking libkrb5-3:amd64 (1.19.2-2ubuntu0.10) over (1.19.2-2ubuntu0.8) ...
Setting up libkrb5-3:amd64 (1.19.2-2ubuntu0.10) ...
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../libgssapi-krb5-2 1.19.2-2ubuntu0.10 amd64.deb ...
Unpacking libgssapi-krb5-2:amd64 (1.19.2-2ubuntu0.10) over (1.19.2-2ubuntu0.8) ...
Setting up libgssapi-krb5-2:amd64 (1.19.2-2ubuntu0.10) ...
Processing triggers for libc-bin (2.35-0ubuntu3.15) ...
root@5f8fe1a0e4ce:/# ^C
root@5f8fe1a0e4ce:/# adduser asaadmin
Adding user `asaadmin' ...
Adding new group `asaadmin' (1000) ...
Adding new user `asaadmin' (1000) with group `asaadmin' ...
Creating home directory `/home/asaadmin' ...
Copying files from `/etc/skel' ...
New password:

```

Figura 5 – Criação do usuário asaadmin e atribuição ao grupo sudo.

Fonte: elaborado pelos autores (2026).

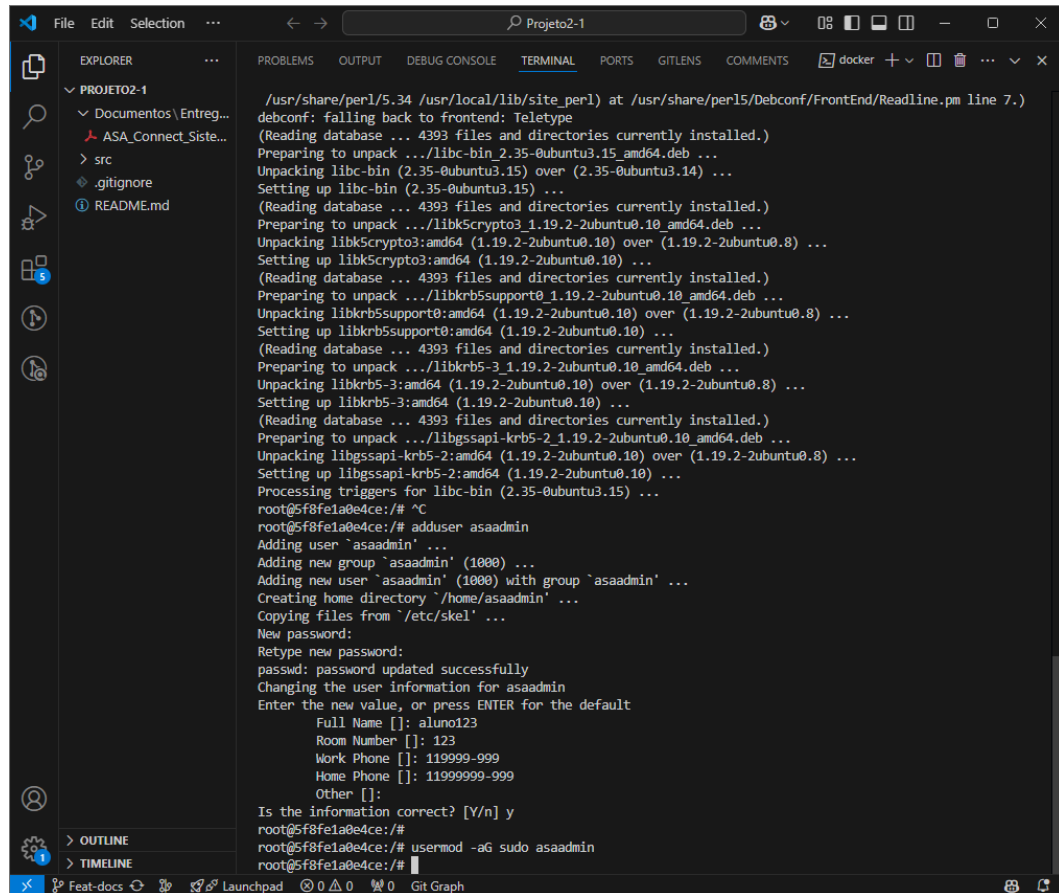
```

File Edit Selection ...
Projeto2-1
EXPLORER
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS COMMENTS
docker + -
v PROJETO2-1
v Documentos\Entreg...
v ASA_Connect_Siste...
> src
> .gitignore
> README.md
> OUTLINE
> TIMELINE
root@5f8fe1a0e4ce:/# ^C
root@5f8fe1a0e4ce:/# adduser asaadmin
Adding user `asaadmin' ...
Adding new group `asaadmin' (1000) ...
Adding new user `asaadmin' (1000) with group `asaadmin' ...
Creating home directory `/home/asaadmin' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for asaadmin
Enter the new value, or press ENTER for the default
Full Name []: alunol23
Room Number []: 123
Work Phone []: 119999-999
Home Phone []: 11999999-999
Other []:
Is the information correct? [Y/n] y
root@5f8fe1a0e4ce:/#

```

Figura 6 – Detalhamento da criação do diretório home do usuário asaadmin.

Fonte: elaborado pelos autores (2026).



```

/usr/share/perl/5.34 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 7.)
debconf: falling back to frontend: Teletype
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../libc-bin 2.35-0ubuntu3.15_amd64.deb ...
Unpacking libc-bin (2.35-0ubuntu3.15) over (2.35-0ubuntu3.14) ...
Setting up libc-bin (2.35-0ubuntu3.15) ...
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../libk5crypto3 1.19.2-2ubuntu0.10_amd64.deb ...
Unpacking libk5crypto3:amd64 (1.19.2-2ubuntu0.10) over (1.19.2-2ubuntu0.8) ...
Setting up libk5crypto3:amd64 (1.19.2-2ubuntu0.10) ...
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../libkrb5support0 1.19.2-2ubuntu0.10_amd64.deb ...
Unpacking libkrb5support0:amd64 (1.19.2-2ubuntu0.10) over (1.19.2-2ubuntu0.8) ...
Setting up libkrb5support0:amd64 (1.19.2-2ubuntu0.10) ...
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../libkrb5-3 1.19.2-2ubuntu0.10_amd64.deb ...
Unpacking libkrb5-3:amd64 (1.19.2-2ubuntu0.10) over (1.19.2-2ubuntu0.8) ...
Setting up libkrb5-3:amd64 (1.19.2-2ubuntu0.10) ...
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../libgssapi-krb5-2 1.19.2-2ubuntu0.10_amd64.deb ...
Unpacking libgssapi-krb5-2:amd64 (1.19.2-2ubuntu0.10) over (1.19.2-2ubuntu0.8) ...
Setting up libgssapi-krb5-2:amd64 (1.19.2-2ubuntu0.10) ...
Processing triggers for libc-bin (2.35-0ubuntu3.15) ...
root@5f8fe1a0e4ce:/# ^C
root@5f8fe1a0e4ce:/# adduser asaadmin
Adding user `asaadmin' ...
Adding new group `asaadmin' (1000) ...
Adding new user `asaadmin' (1000) with group `asaadmin' ...
Creating home directory `/home/asaadmin' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for asaadmin
Enter the new value, or press ENTER for the default
  Full Name []: aluno123
    Room Number []: 123
    Work Phone []: 119999-999
    Home Phone []: 11999999-999
    Other []:
Is the information correct? [Y/n] y
root@5f8fe1a0e4ce:/#
root@5f8fe1a0e4ce:/# usermod -s /bin/bash asaadmin
root@5f8fe1a0e4ce:/#

```

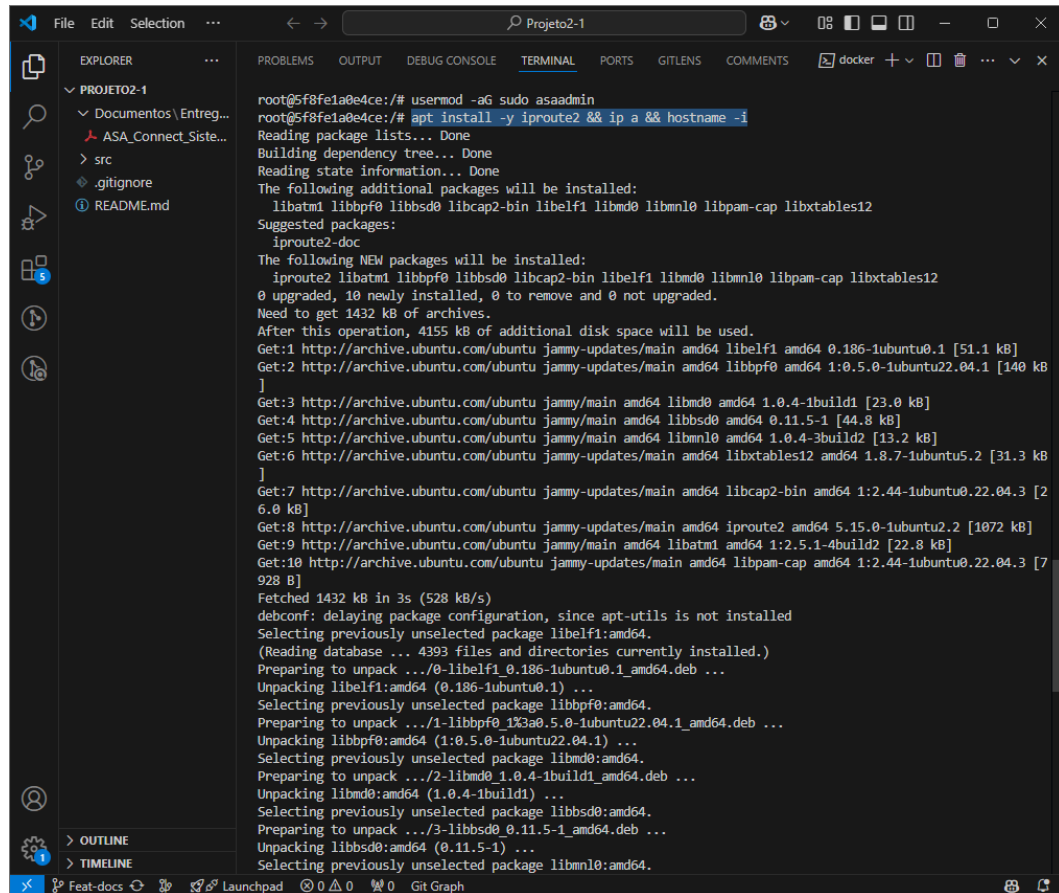
Figura 7 – Confirmação da criação do grupo e do usuário asaadmin (UID/GID 1000).

Fonte: elaborado pelos autores (2026).

2.4 Configuração de Rede

Verificação de interfaces e endereçamento do contêiner:

```
apt install -y iproute2 && ip a && hostname -I
```



```

root@5f8fe1a0e4ce:/# usermod -ag sudo asaadmin
root@5f8fe1a0e4ce:/# apt install -y iproute2 && ip a && hostname -i
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libatm1 libbpf0 libbsd0 libcap2-bin libelf1 libmd0 libmnl0 libpam-cap libxtables12
Suggested packages:
  iproute2-doc
The following NEW packages will be installed:
  iproute2 libatm1 libbpf0 libbsd0 libcap2-bin libelf1 libmd0 libmnl0 libpam-cap libxtables12
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
Need to get 1432 kB of archives.
After this operation, 4155 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libelf1 amd64 0.186-1ubuntu0.1 [51.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libbpf0 amd64 1:0.5.0-1ubuntu22.04.1 [140 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 libmd0 amd64 1.0.4-1build1 [23.0 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/main amd64 libbsd0 amd64 0.11.5-1 [44.8 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 libmnl0 amd64 1.0.4-3build2 [13.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libxtables12 amd64 1.8.7-1ubuntu5.2 [31.3 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcap2-bin amd64 1:2.44-1ubuntu0.22.04.3 [26.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 iproute2 amd64 5.15.0-1ubuntu2.2 [1072 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/main amd64 libatm1 amd64 1:2.5.1-4build2 [22.8 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpam-cap amd64 1:2.44-1ubuntu0.22.04.3 [7928 B]
Fetched 1432 kB in 3s (528 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libelf1:amd64.
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../0-libelf1_0.186-1ubuntu0.1_amd64.deb ...
Unpacking libelf1:amd64 (0.186-1ubuntu0.1) ...
Selecting previously unselected package libbpf0:amd64.
Preparing to unpack .../1-libbpf0_1%3a0.5.0-1ubuntu22.04.1_amd64.deb ...
Unpacking libbpf0:amd64 (1:0.5.0-1ubuntu22.04.1) ...
Selecting previously unselected package libmd0:amd64.
Preparing to unpack .../2-libmd0_1.0.4-1build1_amd64.deb ...
Unpacking libmd0:amd64 (1.0.4-1build1) ...
Selecting previously unselected package libbsd0:amd64.
Preparing to unpack .../3-libbsd0_0.11.5-1_amd64.deb ...
Unpacking libbsd0:amd64 (0.11.5-1) ...
Selecting previously unselected package libmnl0:amd64.
Preparing to unpack .../4-libmnl0_1.0.4-3build2_amd64.deb ...
Unpacking libmnl0:amd64 (1.0.4-3build2) ...
Selecting previously unselected package libxtables12:amd64.
Preparing to unpack .../5-libxtables12_1.8.7-1ubuntu5.2_amd64.deb ...
Unpacking libxtables12:amd64 (1.8.7-1ubuntu5.2) ...
Selecting previously unselected package libcap2-bin:amd64.
Preparing to unpack .../6-libcap2-bin_1:2.44-1ubuntu0.22.04.3_amd64.deb ...
Unpacking libcap2-bin:amd64 (1:2.44-1ubuntu0.22.04.3) ...
Selecting previously unselected package iproute2:amd64.
Preparing to unpack .../7-iproute2_5.15.0-1ubuntu2.2_amd64.deb ...
Unpacking iproute2:amd64 (5.15.0-1ubuntu2.2) ...
Selecting previously unselected package libatm1:amd64.
Preparing to unpack .../8-libatm1_1:2.5.1-4build2_amd64.deb ...
Unpacking libatm1:amd64 (1:2.5.1-4build2) ...
Selecting previously unselected package libpam-cap:amd64.
Preparing to unpack .../9-libpam-cap_1:2.44-1ubuntu0.22.04.3_amd64.deb ...
Unpacking libpam-cap:amd64 (1:2.44-1ubuntu0.22.04.3) ...
Setting up libelf1:amd64 (0.186-1ubuntu0.1) ...
Setting up libbpf0:amd64 (1:0.5.0-1ubuntu22.04.1) ...
Setting up libmd0:amd64 (1.0.4-1build1) ...
Setting up libbsd0:amd64 (0.11.5-1) ...
Setting up libmnl0:amd64 (1.0.4-3build2) ...
Setting up libxtables12:amd64 (1.8.7-1ubuntu5.2) ...
Setting up libcap2-bin:amd64 (1:2.44-1ubuntu0.22.04.3) ...
Setting up iproute2:amd64 (5.15.0-1ubuntu2.2) ...
Setting up libatm1:amd64 (1:2.5.1-4build2) ...
Setting up libpam-cap:amd64 (1:2.44-1ubuntu0.22.04.3) ...
root@5f8fe1a0e4ce:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> <BROADCAST,MULTICAST> mtu 65536
    link/loopback/hwaddr 00:00:00:00:00:00
    inet 127.0.0.1 brd 127.0.0.1 scope host lo
        inet6 ::1::1 scope host
    inet6 ::1::1 scope host
    inet6 fe80::1::1 scope link
    inet6 fe80::1::1 scope link
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> <BROADCAST,MULTICAST> mtu 1500
    link/ether 08:00:27:00:00:00 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.1 brd 10.0.0.1 scope global eth0
        inet6 ::1::1 scope host
    inet6 ::1::1 scope host
    inet6 fe80::1::1 scope link
    inet6 fe80::1::1 scope link
root@5f8fe1a0e4ce:~# hostname -i
10.0.0.1

```

Figura 8 – Saída do comando ip a com as interfaces lo e eth0.

Fonte: elaborado pelos autores (2026).

```

Unpacking libpam-cap:amd64 (1:2.44-1ubuntu0.22.04.3) ...
Setting up libatm1:amd64 (1:2.5.1-4build2) ...
Setting up libcap2-bin (1:2.44-1ubuntu0.22.04.3) ...
Setting up libmnl0:amd64 (1.0.4-3build2) ...
Setting up libxtables12:amd64 (1.8.7-1ubuntu5.2) ...
Setting up libmd0:amd64 (1.0.4-1build1) ...
Setting up libbsd0:amd64 (0.11.5-1) ...
Setting up libelf1:amd64 (0.186-1ubuntu0.1) ...
Setting up libpam-cap:amd64 (1:2.44-1ubuntu0.22.04.3) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/s
hare/perl5/Debconf/FrontEnd/Dialog.pm line 78.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC co
ntains: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.34.0 /usr/local/share/perl/5.34.0 /usr/lib/x86_64-l
inux-gnu/perl5/5.34 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.34
/usr/share/perl/5.34 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 7.)
debconf: falling back to frontend: Teletype
Setting up libpam0:amd64 (1:0.5.0-1ubuntu22.04.1) ...
Setting up iproute2 (5.15.0-1ubuntu2.2) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/s
hare/perl5/Debconf/FrontEnd/Dialog.pm line 78.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC co
ntains: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.34.0 /usr/local/share/perl/5.34.0 /usr/lib/x86_64-l
inux-gnu/perl5/5.34 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.34
/usr/share/perl/5.34 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 7.)
debconf: falling back to frontend: Teletype
Processing triggers for libc-bin (2.35-0ubuntu3.15) ...
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0@if5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 1a:89:d3:5c:1f:b9 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.17.0.2/16 brd 172.17.255.255 scope global eth0
        valid_lft forever preferred_lft forever
172.17.0.2
root@5f8fe1a0e4ce:/#

```

Figura 9 – Endereço IP interno (172.17.0.2/16) atribuído pela bridge padrão do Docker.

Fonte: elaborado pelos autores (2026).

A interface lo (127.0.0.1/8) atende comunicações internas locais; a interface eth0 (172.17.0.2/16) é atribuída pela bridge padrão do Docker, permitindo roteamento entre contêineres e saída externa para APIs (Gemini/OpenAI). A porta 8000/TCP é reservada para a futura API FastAPI (documentação via /docs, OpenAPI/Swagger).

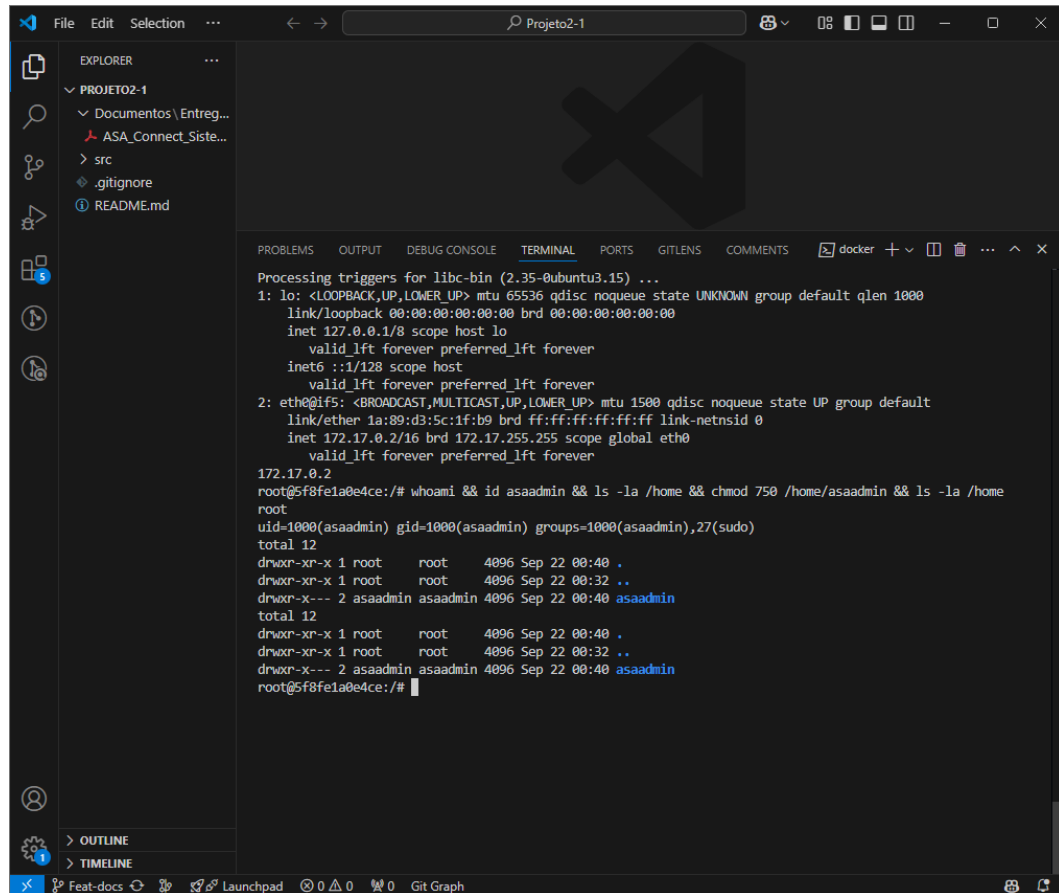
2.5 Permissões e Segurança de Arquivos

Auditoria de identidade e ajuste de permissões do diretório do usuário de aplicação:

```

whoami
id asaadmin
ls -la /home
chmod 750 /home/asaadmin

```



```

Processing triggers for libc-bin (2.35-0ubuntu3.15) ...
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0@if5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 1a:89:d3:5c:1f:b9 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.17.0.2/16 brd 172.17.255.255 scope global eth0
        valid_lft forever preferred_lft forever
172.17.0.2
root@5f8fe1a0e4ce:/# whoami && id asaadmin && ls -la /home && chmod 750 /home/asaadmin && ls -la /home
root
uid=1000(asaadmin) gid=1000(asaadmin) groups=1000(asaadmin),27(sudo)
total 12
drwxr-xr-x 1 root    root    4096 Sep 22 00:40 .
drwxr-xr-x 1 root    root    4096 Sep 22 00:32 ..
drwxr-x--- 2 asaadmin asaadmin 4096 Sep 22 00:40 asaadmin
total 12
drwxr-xr-x 1 root    root    4096 Sep 22 00:40 .
drwxr-xr-x 1 root    root    4096 Sep 22 00:32 ..
drwxr-x--- 2 asaadmin asaadmin 4096 Sep 22 00:40 asaadmin
root@5f8fe1a0e4ce:/#

```

Figura 10 – Saída dos comandos whoami, id, ls -la /home e aplicação da máscara 750.

Fonte: elaborado pelos autores (2026).

A máscara 750 (rwxr-x---) concede ao proprietário (asaadmin) acesso total de leitura, gravação e execução; ao grupo, apenas leitura e execução; e nega qualquer acesso a outros usuários do sistema, em conformidade com os requisitos de privacidade da LGPD.

2.6 Processos, Logs e Consumo de Recursos

Inspeção da tabela de processos isolada pelo namespace PID:

```
ps aux
```

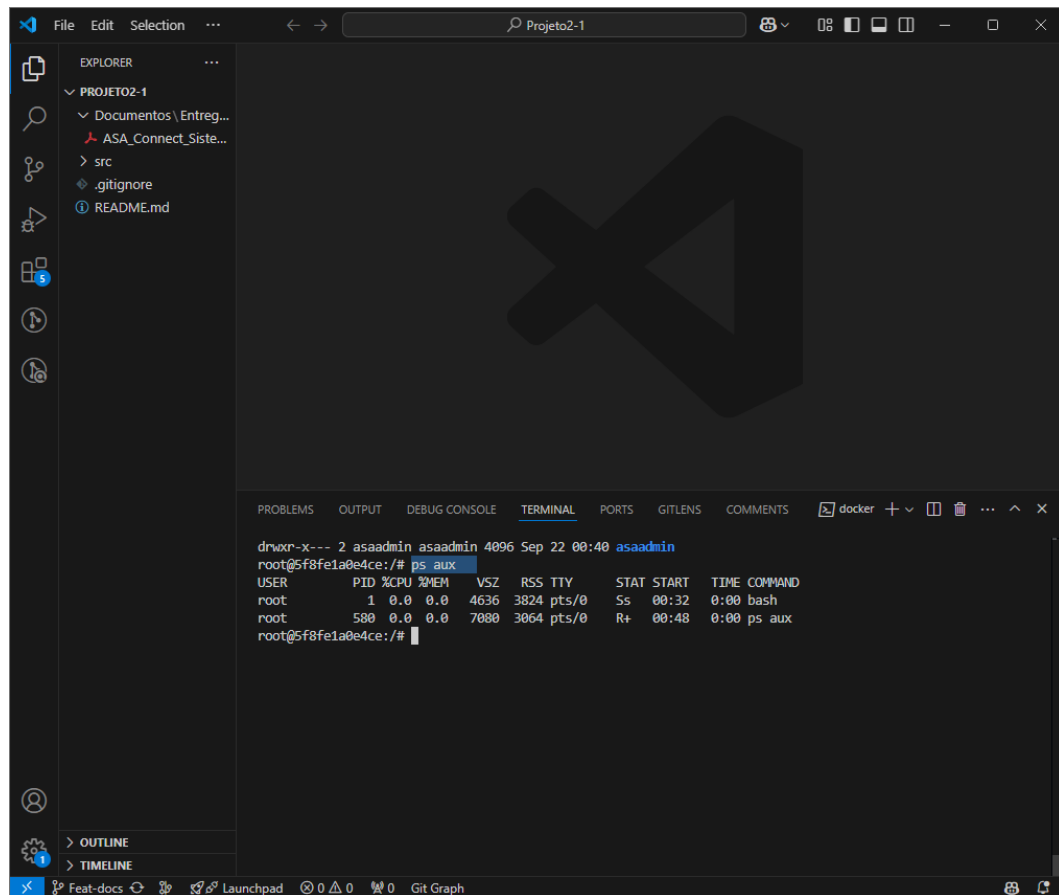


Figura 11 – Listagem de processos em execução (ps aux) — ausência de processos órfãos.

Fonte: elaborado pelos autores (2026).

Verificação dos registros de auditoria e rastreabilidade de instalações (RF10):

```
ls -la /var/log
tail -n 30 /var/log/dpkg.log
```

```

root@5f8fe1a0e4ce:/# ls -la /var/log && tail -n 30 /var/log/dpkg.log
total 348
drwxr-xr-x 1 root root 4096 Sep 1 20:33 .
drwxr-xr-x 1 root root 4096 Sep 1 20:42 ..
-rw-r--r-- 1 root root 4860 Sep 1 20:42 alternatives.log
drwxr-xr-x 1 root root 4096 Sep 22 00:44 apt
-rw-r--r-- 1 root root 64549 Sep 1 20:33 bootstrap.log
-rw-rw---- 1 root utmp 0 Sep 1 20:33 btmp
-rw-r--r-- 1 root root 202870 Sep 22 00:44 dpkg.log
-rw-r--r-- 1 root root 32032 Sep 22 00:40 faillog
-rw-rw-r-- 1 root utmp 292292 Sep 22 00:40 lastlog
-rw-rw-r-- 1 root utmp 0 Sep 1 20:33 wtmp
2026-09-22 00:44:34 status unpacked libxtables12:amd64 1.8.7-1ubuntu5.2
2026-09-22 00:44:34 status half-configured libxtables12:amd64 1.8.7-1ubuntu5.2
2026-09-22 00:44:34 status installed libxtables12:amd64 1.8.7-1ubuntu5.2
2026-09-22 00:44:34 configure libmd0:amd64 1.0.4-1build1 <none>
2026-09-22 00:44:34 status unpacked libmd0:amd64 1.0.4-1build1
2026-09-22 00:44:34 status half-configured libmd0:amd64 1.0.4-1build1
2026-09-22 00:44:34 status installed libmd0:amd64 1.0.4-1build1
2026-09-22 00:44:34 configure libbsd0:amd64 0.11.5-1 <none>
2026-09-22 00:44:34 status unpacked libbsd0:amd64 0.11.5-1
2026-09-22 00:44:34 status half-configured libbsd0:amd64 0.11.5-1
2026-09-22 00:44:34 status installed libbsd0:amd64 0.11.5-1
2026-09-22 00:44:34 configure libelf1:amd64 0.186-1ubuntu0.1 <none>
2026-09-22 00:44:34 status unpacked libelf1:amd64 0.186-1ubuntu0.1
2026-09-22 00:44:34 status half-configured libelf1:amd64 0.186-1ubuntu0.1
2026-09-22 00:44:34 status installed libelf1:amd64 0.186-1ubuntu0.1
2026-09-22 00:44:34 configure libpam-cap:amd64 1:2.44-1ubuntu0.22.04.3 <none>
2026-09-22 00:44:34 status unpacked libpam-cap:amd64 1:2.44-1ubuntu0.22.04.3
2026-09-22 00:44:34 status half-configured libpam-cap:amd64 1:2.44-1ubuntu0.22.04.3
2026-09-22 00:44:34 status installed libpam-cap:amd64 1:2.44-1ubuntu0.22.04.3
2026-09-22 00:44:34 configure libbpf0:amd64 1:0.5.0-1ubuntu22.04.1 <none>
2026-09-22 00:44:34 status unpacked libbpf0:amd64 1:0.5.0-1ubuntu22.04.1
2026-09-22 00:44:34 status half-configured libbpf0:amd64 1:0.5.0-1ubuntu22.04.1
2026-09-22 00:44:34 status installed libbpf0:amd64 1:0.5.0-1ubuntu22.04.1
2026-09-22 00:44:34 configure iproute2:amd64 5.15.0-1ubuntu2.2 <none>
2026-09-22 00:44:34 status unpacked iproute2:amd64 5.15.0-1ubuntu2.2
2026-09-22 00:44:34 status half-configured iproute2:amd64 5.15.0-1ubuntu2.2
2026-09-22 00:44:35 status installed iproute2:amd64 5.15.0-1ubuntu2.2
2026-09-22 00:44:35 trigproc libc-bin:amd64 2.35-0ubuntu3.15 <none>
2026-09-22 00:44:35 status half-configured libc-bin:amd64 2.35-0ubuntu3.15
2026-09-22 00:44:35 status installed libc-bin:amd64 2.35-0ubuntu3.15
root@5f8fe1a0e4ce:/#

```

Figura 12 – Listagem de /var/log e últimas entradas do dpkg.log.

Fonte: elaborado pelos autores (2026).

Monitoramento de CPU, memória e disco:

```

top -b -n 1
free -h
df -h

```

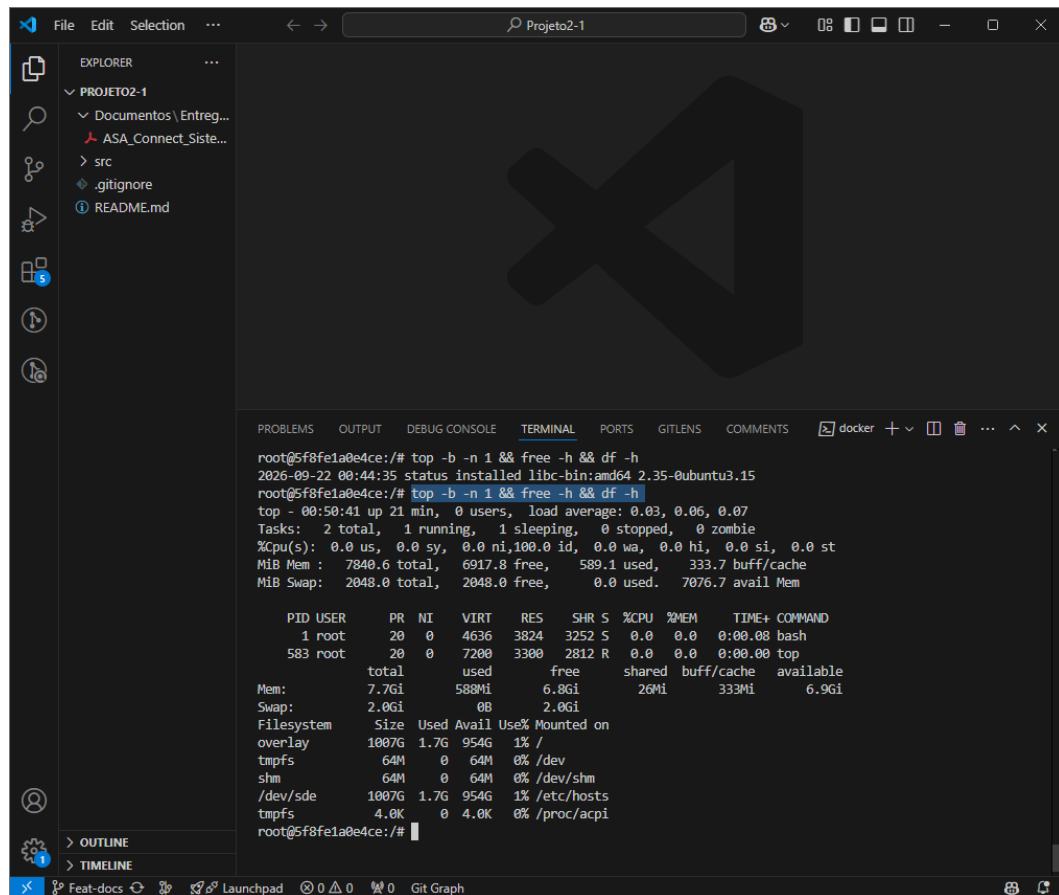


Figura 13 – Consumo de CPU (~0% ocioso), memória (< 600 MB) e disco (overlay2, 1%).

Fonte: elaborado pelos autores (2026).

2.7 Diagrama da Infraestrutura

A arquitetura do ecossistema ASA Connect integra quatro camadas: apresentação (app Flutter), contêiner Docker com a API FastAPI (autenticação JWT, pipeline RAG por similaridade de cosseno/TF-IDF e módulo de auditoria), persistência (base de conhecimento indexada e banco relacional PostgreSQL/SQLite) e serviço externo de inferência (Google Gemini LLM).

Fluxo de dados:

- 1) O estudante autentica-se via token JWT no app Flutter e envia requisição HTTPS/REST à porta 8000 do contêiner.
- 2) O pipeline RAG consulta a base de conhecimento oficial do ASA e monta o prompt enriquecido.
- 3) A requisição é encaminhada à API do Google Gemini via canal TLS, com a chave recuperada de variável de ambiente não versionada.
- 4) A resposta sintetizada retorna com citação de fontes; cada evento é registrado em log e no banco de dados para auditoria.

2.8 Política Básica de Segredos

- Vedação de credenciais no versionamento: GEMINI_API_KEY, JWT_SECRET_KEY e POSTGRES_PASSWORD nunca são commitadas no Git.
- Arquivo .env.example documenta as variáveis exigidas com valores fictícios, sem expor dados sensíveis.
- O .gitignore bloqueia o envio acidental de .env, *.pem, *.key, *.db e __pycache__.
- Todos os serviços executam sob o usuário asaadmin, sem privilégios de root, em conformidade com o menor privilégio.

2.9 Evidências e Repositório

Dockerfile e docker-compose.yml em /src/backend; variáveis de ambiente em /src/backend/.env.example; regras de proteção de segredos em /.gitignore; módulo do agente e base de conhecimento em /src/backend/app/agent/ e /src/backend/knowledge_base/.
Repositório oficial: <https://github.com/2026-2-NCC5/Projeto2>.

3 CONCLUSÃO

A Entrega 1 comprovou o provisionamento e a validação bem-sucedidos do ambiente operacional Linux em contêiner Docker (Ubuntu 22.04 LTS), com baixo footprint computacional (CPU próxima a 0% em repouso; RAM inferior a 600 MB) e aderência às boas práticas de segurança: usuário não-root (asaadmin), permissões restritivas (750), isolamento por namespaces/cgroups e governança de segredos via .env.example e .gitignore. O ambiente está apto a suportar a Entrega 2, que contemplará o deploy em nuvem pública (Render/AWS), a orquestração da API FastAPI com PostgreSQL via Docker Compose, o monitoramento de métricas em tempo real (latência, taxa de erro, throughput), testes de carga com acessos concorrentes e a integração de ponta a ponta com a aplicação Flutter.